

Esercizio 5

(I)

$$\frac{\frac{Ax}{\Gamma \vdash p \Rightarrow \sigma \Rightarrow \tau} \quad \frac{Ax}{\Gamma \vdash p}}{\Gamma \vdash \sigma \Rightarrow \tau} \Rightarrow e$$

$$\frac{\frac{Ax}{\Gamma \vdash p \Rightarrow \sigma} \quad \frac{Ax}{\Gamma \vdash p}}{\Gamma \vdash \sigma} \Rightarrow e$$

$$\frac{\Gamma = (p \Rightarrow \sigma \Rightarrow \tau), (p \Rightarrow \sigma), p \vdash \tau}{(p \Rightarrow \sigma \Rightarrow \tau), (p \Rightarrow \sigma) \vdash p \Rightarrow \tau} \Rightarrow e$$

$$\frac{(p \Rightarrow \sigma \Rightarrow \tau), (p \Rightarrow \sigma) \vdash p \Rightarrow \tau}{(p \Rightarrow \sigma \Rightarrow \tau) \vdash (p \Rightarrow \sigma) \Rightarrow p \Rightarrow \tau} \Rightarrow i$$

$$\frac{(p \Rightarrow \sigma \Rightarrow \tau) \vdash (p \Rightarrow \sigma) \Rightarrow p \Rightarrow \tau}{\vdash (p \Rightarrow \sigma \Rightarrow \tau) \Rightarrow (p \Rightarrow \sigma) \Rightarrow p \Rightarrow \tau} \Rightarrow i$$

$$\vdash (p \Rightarrow \sigma \Rightarrow \tau) \Rightarrow (p \Rightarrow \sigma) \Rightarrow p \Rightarrow \tau$$

(II)

$$\frac{\frac{Ax}{\Gamma \vdash p \Rightarrow \perp} \quad \frac{Ax}{\Gamma \vdash p}}{\Gamma = p \Rightarrow \perp, p \vdash \perp} \Rightarrow e$$

$$\frac{p \Rightarrow \perp \vdash \neg p}{\vdash (p \Rightarrow \perp) \Rightarrow \neg p} \neg i$$

$$\vdash (p \Rightarrow \perp) \Rightarrow \neg p$$

(III)

$$\frac{\frac{Ax}{p, \neg p \vdash p} \quad \frac{Ax}{p, \neg p \vdash \neg p}}{p, \neg p \vdash \perp} \neg e$$

$$\frac{p, \neg p \vdash \perp}{p \vdash \neg \neg p} \neg i$$

$$\frac{p \vdash \neg \neg p}{p \Rightarrow \neg \neg p} \Rightarrow i$$

IV

$$\begin{array}{c}
 \frac{Ax}{\Gamma \vdash p} \quad \frac{Ax}{\Gamma \vdash \neg p} \\
 \hline
 \Gamma, \neg p \vdash \perp \quad \neg e \\
 \neg i \quad \frac{\Gamma \vdash \neg p \quad \Gamma \vdash \neg \neg p}{\Gamma \vdash \neg p, p \vdash \perp} \neg e \\
 \hline
 \Gamma \vdash \neg p \vdash p \\
 \hline
 \Gamma \vdash \neg p \Rightarrow p \quad \Rightarrow i
 \end{array}$$

V

$$\begin{array}{c}
 \frac{Ax}{\Gamma \vdash p \Rightarrow \sigma} \quad \frac{Ax}{\Gamma \vdash p} \\
 \hline
 \Gamma \vdash \sigma \quad \Rightarrow e \quad \frac{Ax}{\Gamma \vdash \neg \sigma} \\
 \hline
 \Gamma \vdash p \Rightarrow \sigma, \neg \sigma, p \vdash \perp \quad \neg e \\
 \hline
 \Gamma \vdash p \Rightarrow \sigma, \neg \sigma \vdash p \quad \neg i \\
 \hline
 \Gamma \vdash p \Rightarrow \sigma \vdash \neg \sigma \Rightarrow p \quad \Rightarrow i \\
 \hline
 \Gamma \vdash (p \Rightarrow \sigma) \Rightarrow (\neg \sigma \Rightarrow p) \quad \Rightarrow i
 \end{array}$$

VI

$$\begin{array}{c}
 \frac{Ax}{\Gamma \vdash (p \wedge \sigma) \Rightarrow \tau} \quad \frac{Ax}{\Gamma \vdash p} \quad \frac{Ax}{\Gamma \vdash Q} \\
 \hline
 \Gamma \vdash (p \wedge \sigma) \Rightarrow \tau, p \wedge Q \quad \wedge i \\
 \hline
 \Gamma \vdash ((p \wedge \sigma) \Rightarrow \tau), p, \sigma \vdash \tau \quad \Rightarrow e \\
 \hline
 \Gamma \vdash ((p \wedge \sigma) \Rightarrow \tau), p \vdash \sigma \Rightarrow \tau \quad \Rightarrow i \\
 \hline
 \Gamma \vdash ((p \wedge \sigma) \Rightarrow \tau) \vdash (p \Rightarrow \sigma \Rightarrow \tau) \quad \Rightarrow i \\
 \hline
 \Rightarrow ((p \wedge \sigma) \Rightarrow \tau) \Rightarrow (p \Rightarrow \sigma \Rightarrow \tau) \quad \Rightarrow i
 \end{array}$$

$$\begin{array}{c}
\frac{\frac{Ax}{\Gamma \vdash p \Rightarrow \sigma \Rightarrow \tau}}{\Gamma \vdash \sigma \Rightarrow \tau} \Rightarrow e \quad \frac{\frac{Ax}{\Gamma \vdash p \wedge \sigma}}{\Gamma \vdash p} \wedge e_1 \quad \frac{Ax}{\Gamma \vdash p \wedge \sigma} \wedge e_2 \\
\frac{\Gamma \vdash (p \Rightarrow \sigma \Rightarrow \tau), (p \wedge \sigma) \vdash \tau}{(p \Rightarrow \sigma \Rightarrow \tau) \vdash ((p \wedge \sigma) \Rightarrow \tau)} \Rightarrow e \\
\frac{(p \Rightarrow \sigma \Rightarrow \tau) \vdash ((p \wedge \sigma) \Rightarrow \tau)}{(p \Rightarrow \sigma \Rightarrow \tau) \Rightarrow ((p \wedge \sigma) \Rightarrow \tau)} \Rightarrow i \\
\Leftarrow
\end{array}$$